

(a) the substrate = any per-token attention KV

attention variant	status
FlashAttn / paged / vLLM	free
GQA / MQA	free
MLA (DeepSeek-V2/V3)	adapter
M-RoPE (sectioned/interleaved)	adapter
Sliding-window (Gemma)	fixed
Hybrid attn+SSM (Falcon-H1)	partial
Seq.-dim compress (V4 CSA/HCA)	open
RWKV / Mamba / diffusion	out

(b) images are position-portable too

